MIL-S-12883/9B
11 July 1968
SUPERSEDING
MIL-S-12883/9A
12 June 1961
MS91513
5 October 1954

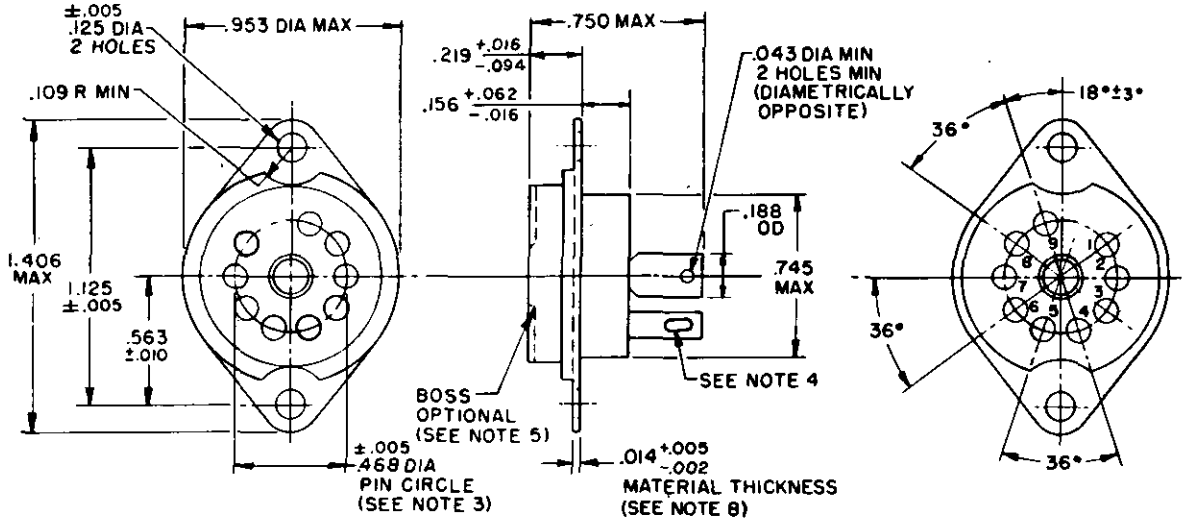
MILITARY SPECIFICATION SHEET

SOCKETS AND ACCESSORIES FOR PLUG-IN ELECTRONIC COMPONENTS (ELECTRON TUBE, TOP MOUNTING, SADDLE TYPE WITHOUT SHIELD BASE, 9 CONTACT, RADIAL)

INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION SHEET. USE MIL-S-12883/11.

The complete requirements for procuring the sockets and accessories described herein shall consist of this document and the latest issue of Specification MIL-S-12883.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.043	1.09	.156	3.96	.745	18.92
.005	.13	.062	1.57	.188	4.78	.750	19.05
.010	.25	.094	2.39	.219	5.56	.953	24.21
.014	.36	.109	2.77	.468	11.89	1.125	28.58
.016	.41	.125	3.18	.563	14.30	1.406	35.71

NOTES:
1. All dimensions in inches. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Unless otherwise specified, tolerances are .xxx±.016 (.41 mm) and ±1/2° on angles.
3. Nine contact cavities equally spaced throughout 288° of arc, and each cavity located within 1/2° of true position, shall be established along the pin circle.
4. Each contact tab shall have either 1 wire hole of .040 (1.02 mm) minimum width and .125 (3.18 mm) minimum length, or 2 holes of .040 (1.02 mm) minimum width and .075 (1.91 mm) minimum length. The hole, or holes, shall lie on the longitudinal centerline of the contact tab within ±.008 (.20 mm).
5. The design of bosses and the shape of barriers, when used, are optional. When present, barriers shall have a maximum height of .062 (1.57 mm).
6. Clearance between the cavity wall and the contact, with the contact in any position, shall be no greater than .018 (.46 mm).
7. Marking shall be at any convenient visible location.
8. Material thickness of mounting flange shall be measured in an area where burring or dishing of the mounting hole is not present.

FIGURE 1. Socket configuration.

FSC 5935

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REQUIREMENTS:

Dimensions, configuration, and descriptive information: See figure 1 and table I.

Contact cavities: One shape only; D-shaped or circular, at the option of the supplier.

Float: With a pin of 0.041-inch diameter and 0.271-inch minimum length fully inserted in a socket contact, the contact shall be capable of free movement (float) within the contact cavity.

Saddle: The method of attaching the saddle to the insulator body, and the shape of the saddle where attachment is effected, are optional.

Insertion and withdrawal force:
Initial insertion force: 20 pounds maximum.
Average insertion force: 15 pounds maximum.
Initial withdrawal force: 4 pounds minimum; 15 pounds maximum.

Insulation resistance: 1,000 megohms minimum.
(Test-pin diameter: 0.040 ±0.001 inch.)

Dielectric withstanding voltage:
Sea level: Test voltage: 2,000 volts root mean square (rms).
(Test-pin diameter: 0.040 ±0.001 inch.)
High altitude: Test voltage: 660 volts rms.
(Test-pin diameter: 0.040 ±0.001 inch.)

Contact resistance:
Average for all contacts: 0.015 ohm maximum.
Individual contacts: 0.03 ohm maximum.

Vibration: The test-gage shall be an approved electron-tube type 12AT7WA, in accordance with MIL-E-1/3.

Continuity test circuit: With the header test gage(s) inserted into the socket(s) under test, the pins of the test gage(s) and the contacts of the socket(s) under test shall result in a series circuit. The header shall be the one used for electron tube 12AT7WA.

Durability: After the durability test, the withdrawal force shall be 2 pounds minimum.

Static load: 52 pounds.

Test-gage details: See table II and MIL-S-12883.

Mating-base dimensions: Sockets shall accommodate plug-in components having mating-base dimensions as shown on figure 2.

TABLE I. Dash numbers, military part numbers, and descriptive information.

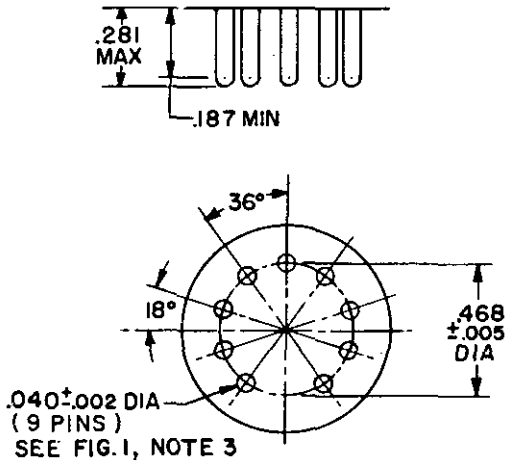
Dash number	Military part number	Operating temperature °C	Insulating material	Old type designator and replacement data 1/
01	M12883/09-01	200	Ceramic	TS103C02 (2-way)
02	M12883/09-02	100	SDG-F or GDI-30F	TS103P02 (1-way)
03	M12883/09-03	100	MIL-M-14, type MFE	TS103P02 (2-way)

1/ 1-way replacement: The new item (formerly designated by military part number) will replace the old item (formerly designated by type designator); however, the old item will not replace the new item.
2-way replacement: The two items are interchangeable; i.e., the new item will replace the old item, and the old item will replace the new item.

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TABLE II. Test-gage details.

Inspection	A (pin length)	B (pin dia)	M (test-end dia)	C (pin-circle dia)	N (probe-end dia)	D (gage dia) (max)	No. of pins	Total weight of gage (±5%)
	Inch	Inch	Inch	Inch	Inch	Inch		Ounces
Insertion and withdrawal force - - - -	0.276 ±0.005	0.0390 ±0.0001	---	0.4680 basic	---	47/64	9	---
Contact resistance -	0.276 ±0.005	0.0390 ±0.0005	---	0.4680 basic	---	47/64	9	---
Contact retention - -	0.276 ±0.005	---	0.0390 ±0.0001	---	0.0410 ±0.0001	---	-	3
Durability - -	0.276 ±0.005	0.0410 ±0.0001	---	0.4680 basic	---	47/64	9	---



INCHES	MM
.002	.05
.005	.13
.040	1.02
.187	4.75
.281	7.14
.468	11.89

NOTE:
All dimensions in inches. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

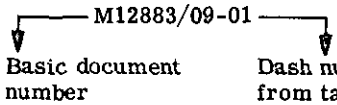
FIGURE 2. Mating-base dimensions.

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OPERATING INFORMATION:

Ratings (absolute maximum):
Voltage:
Sea level: 500 volts rms.
50,000 ft: 225 volts rms.
Current: 1 ampere.
Operating temperature: See table I.

Part number: Consists of the basic number of this specification sheet and a dash number from table I.
Example:



Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - MI, MU, EL, ME, AT, AV
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - None
Navy - MC, AS, OS, CG
Air Force - 19

Preparing activity:
Army - EL

(Project 5935-1357-8)

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